MIL-C-82660(OS)
31 January 1977
Superseding
(See Section 6)

MILITARY SPECIFICATION

CHROMIUM 2-ETHYL HEXOATE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of chromium 2-ethyl hexoate.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on data of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipping and Storage
MIL-STD-1218	ACS Chemicals

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Benefical comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station (611), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of the invitation for bids or the request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 70-72 Specific Gravity of Semi-solid Bituminous Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

CODE OF FEDERAL REGULATIONS

49 CFR 100-199 Transportation

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. Orders should specify "49 CFR 100-199 (latest revision)".)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section 1616 P Street, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference form libraries. They are distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The material shall be a dark green liquid that is essentially pure chromium 2-ethyl hexoate diluted with 2-ethyl hexanoic acid.

3.2 Chemical and physical properties. The chemical and physical properties shall be in accordance with TABLE I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.

TABLE I. Chemical and Physical Properties.

	Values	
	minimum	maximum
Chromium content, wt%	5.3	5.7
Specific gravity, 25/25°C	0.97	1.03

3.3 Workmanship. The material shall be uniform, free from contamination, foreign material or any other defect that would prevent its use for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 18 to 35°C (65 to 95°F)

b. Altitude: Normal ground

c. Vibration: None

d. Humidity: Room ambient to 95 percent relative, maximum

4.3 Sampling.

4.3.1 Lot. Unless otherwise specified in the contract (see 6.2), a lot shall consist of all material manufactured in one continuous production run or in one batch, under essentially identical conditions, from the same raw materials, and to be offered for acceptance at one time. Several batches, manufactured from the same raw materials, may be blended to form a uniform larger batch which shall then consititute a lot for inspection purposes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.

4.3.2 Sampling. Sampling for quality conformance inspection shall be in accordance with inspection Level I of MIL-STD-105. The sample unit shall be one unit package or container of material. Each sample shall consist of sufficient material to perform the quality conformance tests as specified in 4.4.

4.4 Quality conformance inspection. Each sample obtained in accordance with 4.3.2 shall be subjected to the tests and examination of 4.5. The acceptable quality level (AQL) shall be 2.5% defective. When specified in the contract (see 6.2), the contractor shall furnish test reports showing quantitative results for all quality conformance tests specified for each lot of material.

4.5 Tests and examinations. Unless otherwise specified herein, all chemicals shall be ACS grade in accordance with MIL-STD-1218.

4.5.1 Visual examination. All samples shall be visually examined to verify conformance to the workmanship requirement.

4.5.2 Chromium content. Chromium content shall be determined in accordance with the following:

a. Reagents:

1. Sulfuric acid.
2. Approximately 5 percent sulfuric acid solution.
3. Sulfuric acid diluted 1:1 by volume.
4. Nitric acid.
5. Approximately 50% sodium hydroxide solution.
6. Sodium peroxide.
7. o-Phosphoric acid, 85%
8. Ferrous ammonium sulfate.
9. Standardized 0.1 normal (N) potassium dichromate; 4.9035 grams/liter (g/l).
10. Diphenylamine sulfonate indicator, 0.2% (dissolve 0.2 g p-diphenylamine sulfonic acid sodium salt in 100 ml of water.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.

b. Apparatus:

1. Erlenmeyer flask, 300 ml.

2. Hot plate.

3. Bunsen or Meeker burner.

4. Beaker, 600 ml.

5. Buret, 50 ml.

6. Magnetic stirrer with teflon coated stirring bar.

7. Volumetric flask, 1 liter.

c. Standardization of 0.1 N ferrous ammonium sulfate: Add 39.22 g of ferrous ammonium sulfate to a one liter volumetric flask and dissolve in 5% sulfuric acid. Dilute to volume with 5% sulfuric acid. Standardize with 0.1 N potassium dichromate solution with diphenylamine sulfonate indicator to a blue-green endpoint.

d. Procedure: Weigh a 0.48 to 0.52 g sample, to the nearest 0.1 mg, on a small piece of filter paper. Add paper and sample to a 300 ml Erlenmeyer flask. Add 5 ml concentrated sulfuric acid and 10 ml concentrated nitric acid to the flask and digest on a hot plate until all organic material is destroyed. Repeat dropwise addition of nitric acide until a clear green solution is obtained. Heat to slight fumes of sulfuric acid. (Care must be taken in this heating as excessive fuming with sulfuric acid produces insoluble chromium salts.) Cool to 30 ± 5°C and add 25 to 50 ml of water slowly down the side of the flask to avoid spattering. Add sufficient 50% sodium hydroxide solution to the flask until the mixture is alkaline to pH paper. Add 2 g of sodium peroxide and dilute to 100 ml with water. Boil gently to a volume of 40 ml to remove residual sodium peroxide. Cool and transfer quantitatively to a 600 ml beaker and dilute to a volume of 300 ml. Add 5 ml phosphoric acid and 50 ml 1:1 sulfuric acid. Titrate with 0.1 N ferrous ammonium sulfate to a blue green endpoint using 3 drops of diphenylamine sulfonate indicator.

e. Calculation: Chromium content, wt% $= \left[\frac{0.01734\ (A)(N)}{B}\right] x\ 100$

where: A = volume of standardized ferrous ammonium sulfate, ml
N = normality of standardized ferrous ammonium sulfate
B = weight of sample, g
0.01734 = factor to convert from ferrous ammonium sulfate to elemental chromium

f. Report the results of a minimum of 2 determinations and their average.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.

4.5.3 Specific gravity. Specific gravity, 25/25°C, shall be determined in accordance with ASTM D 70-72.

4.6 Packaging inspection. The packaging, packing and marking shall be inspected to verify conformance with the requirements of Section 5.

5. PACKAGING

5.1 Packaging and packing. Unles otherwise specified in the contract (see 6.2), packaging and packing shall be Level C.

5.1.1 Level C. Unless otherwise specified in the contract (see 6.2), packaging and packing of chromium 2-ethyl hexoate shall be in accordance with standard commercial practice applicable to the type of material. The packaging and packing shall be of such construction and materials that the contents will be adequately protected against loss or contamination. Container size shall be as specified in the contract (see 6.2). Containers shall conform to Uniform Freight Classification, National Motor Freight Classification or to rules of other carriers applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract (see 6.2), each container shall be marked in accordance with MIL-STD-129 and Code of Federal Regulations 49 CFR 171-179. Marking shall include, but not be limited to, the following information:

a. Title, number and date of this specification
b. Manufacturer's name and location
c. Material trade name
d. Net weight and volume
e. Lot number, batch number(s), and date of manufacture
f. Storage conditions
g. Contract or purchase order number

6. NOTES

6.1 Intended use. Chromium 2-ethyl hexoate in accordance with this specification is intended for use as an ingredient in solid propellant for the CCU-22/A impulse cartridge and other cartridge devices.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.

a. Title, number and date of this specification
b. Quantity required
c. Place of delivery
d. Inspection conditions when other than as specified (see 4.2)
e. Lot size if other than as specified (see 4.3.1)
f. Packaging requirements if other than as specified (see 5.1)
g. Size of container required (see 5.1.1)
h. Any special markings required (see 5.2)

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DID*
4.4	Quality conformance inspection data	-

*DIDs (Data Item Description/DD Form 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDs when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

6.3 Supersession information. MIL-C-82660 is intended to be used in lieu of AS 3059 (Code Ident 30003) dated 6 December 1974 for Naval Sea Systems Command procurement.

6.4 Suggested source of supply. A product that has met the requirements of this specification in past procurement actions is chromium 2-ethyl hexoate, 5.5% Cr, manufactured by Shepherd Chemical Company, 4900 Beech St., Cincinnati, OH 45212. This information is for the convenience of the procuring activity and is not to be construed as a waiver of any requirement of this specification nor as any limitation of additional potential sources of supply.

Custodian:
Navy - OS

Preparing Activity:
Navy - OS
Project Number:
6810-NA34

☆ U. S. GOVERNMENT PRINTING OFFICE: 1977-703-020/1927

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.